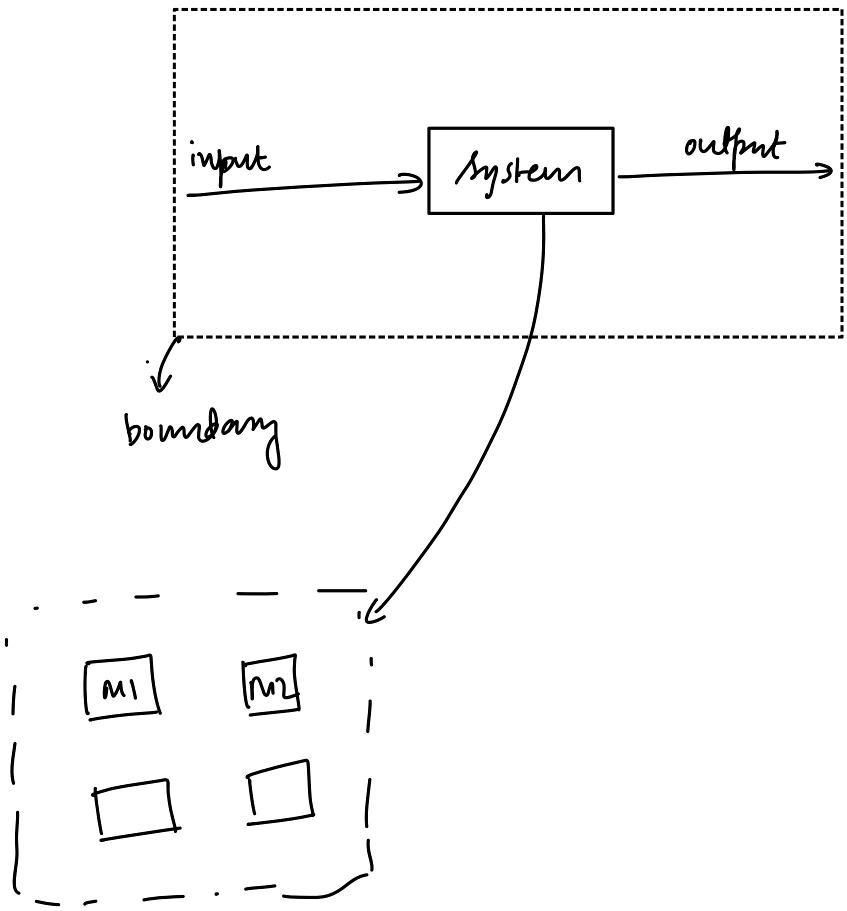
Book: ① Model Checking by Edmund Clarke

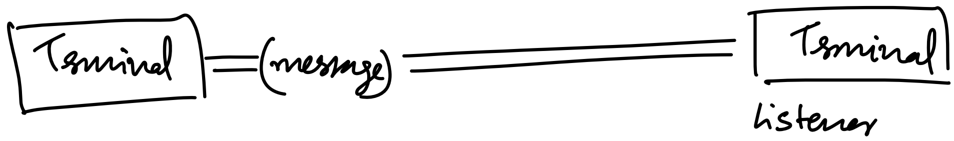
② Principles of Model Checking Christel Baier & Joost-Pieter Katoen.

Systems: Basics:



Environment
- → people
- → component
- → Network

MSE	20
ESE	40
Internal Evaluation	30
AH	10



Isolated Closed Open

What is a software system

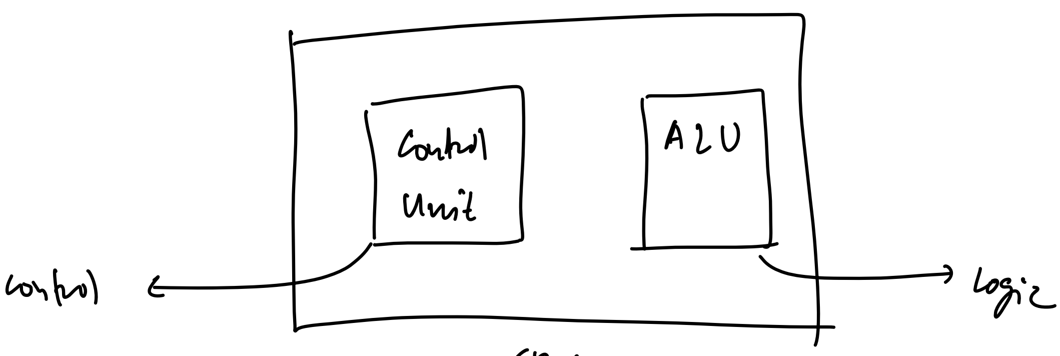
→ core functionality should be in the software

SRS document

- → requirement 1
- → req. 2
- → req. n

waterfall model

Program = logic + control + data structures



CPU

Software Systems from a Verification Perspective

→ Operate in logical (digital) environments

DCU / MCU / CPU / GPU

→ Interact through API's

→ Fail in logical ways

Verification focuses on:

- functional correctness
  - ↪ non-function properties, like temporal properties
    - e.g. module should respond with this value within this time
-
-

Software Intensive Systems

- Eg1: ERP Systems
- Eg2: HIMS

Software as medical devices (SAMD)

Embedded Systems

>> 99% → Embedded Systems

<< 1% → GP deploy

$\underbrace{\hspace{6em}}_{\text{GPU}}$

Software Intensive Systems vs Embedded Systems ?

control program

main() {

Discrete

}

physical world

Temperature

Humidity

Temp $\in [20, 23]$

Humidity $\in [\ ,\ ]$

we have variables
that are continuous (double values)

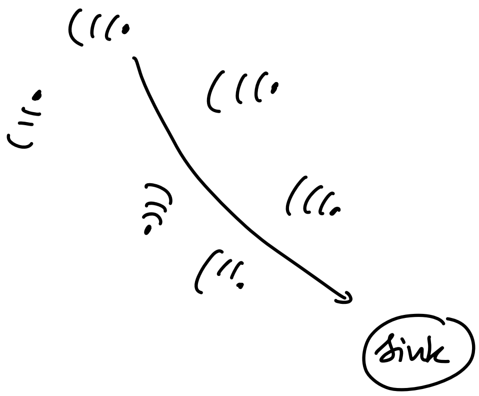
infinitely precise

finitely precise
IEEE restrictions

Wireless Sensor Networks (WSN)

(Sensors / Actuators)

→ have a large number of nodes



Internet of Things (IoT)

→ Internet connectivity via IP addresses.

Knowledge About the System matters in verificati

1. Software systems
2. Software intensive systems
3. Embedded systems
4. CPS
5. WSN
6. IoT